

Q4) solve the following algorithm with Round Robin algorithm. using time quantum of 2 & context switch time of 0.5...

<u>Process</u>	<u>AT</u>	<u>BT</u>
P ₁	0	5
P ₂	0	4
P ₃	0	3

Sol

Ready Queue (RQ)

t = 0

RQ = [P₁ | P₂ | P₃] → P₁ gets 2 units
 P₁ :- 5 - 2 → 3
 Not completed → move back

t = 2

RQ = [P₂ | P₃ | P₁] → CS = 0.5

t = 2.5

RQ = [P₂ | P₃ | P₁] → P₂ gets 2 units
 P₂ :- 4 - 2 = 2
 Not completed → move back

t = 4.5

RQ = [P₃ | P₁ | P₂] → CS = 0.5

$$\underline{t = 5}$$

RQ =

P_3	P_1	P_2
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 $\rightarrow P_3$ gets 2 units
 $P_3: 3 - 2 = 1$
 Not completed $\rightarrow$ move back

$$\underline{t = 7}$$

RQ =

P_1	P_2	P_3
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 $\rightarrow CS = 0.5$

$$\underline{t = 7.5}$$

RQ =

P_1	P_2	P_3
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 $\rightarrow P_1$ gets 2 units
 $P_1: 3 - 2 \rightarrow 1$
 not completed $\rightarrow$ move back

$$\underline{t = 9.5}$$

RQ =

P_2	P_3	P_1
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 $\rightarrow CS = 0.5$

$$\underline{t = 10}$$

RQ =

P_2	P_3	P_1
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 $\rightarrow P_2$ gets 2 units
 $P_2: 2 - 2 = 0$
 P_2 completed.

$$\underline{t = 12}$$

$$CS = 0.5$$

$$\underline{t = 12.5}$$

RQ =

P_3	P_1
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 $\rightarrow P_3$ gets 1 unit
 $P_3: 1 \rightarrow 0$
 P_3 Completed

$$t = 13.5$$

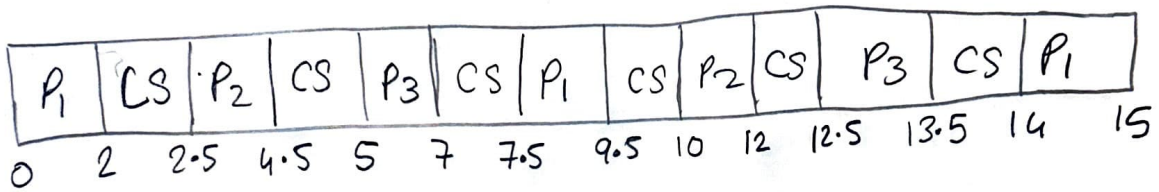
$$CS = 0.5$$

$$t = 14$$

$RQ = \boxed{P_1}$ $\rightarrow P_1$ gets 1 unit
 $P_1: 1 \rightarrow 0$
 P_1 completed

$t = 15 \rightarrow RQ$ is empty.

Gantt chart (GT)



<u>Process</u>	<u>AT</u>	<u>BT</u>	<u>CT</u>	<u>TAT</u>	<u>WT</u>	<u>RT</u>
P_1	0	5	15	15	10	0
P_2	0	4	12	12	8	2.5
P_3	0	3	13.5	13.5	10.5	5

$$\Rightarrow \underline{\text{Avg WT}} = \underline{9.5} \quad \Rightarrow \underline{\text{Total elapsed time}} = \underline{15}$$

$$\Rightarrow \underline{\text{Avg TAT}} = \underline{13.5} \quad \Rightarrow \underline{\text{Through Put}} = \underline{0.2 \text{ Processes/unit}}$$

$$\Rightarrow \underline{\text{Avg RT}} = \underline{2.5}$$